SE Lab 5
Reports Screenshots

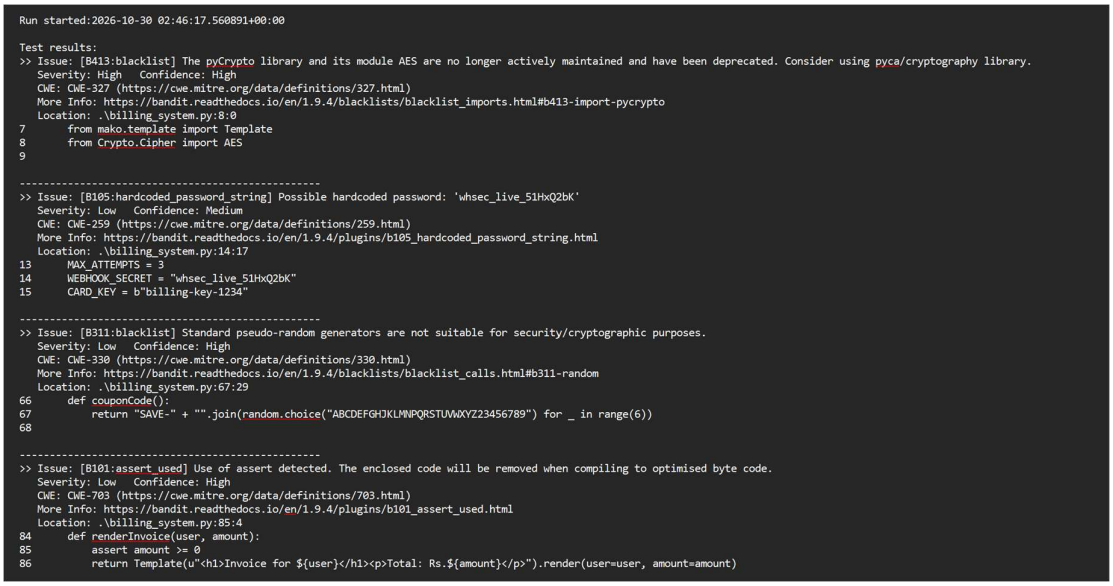
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SRN: PES1UG24CS909

Class: 5G

Before:

1) bandit_report.txt



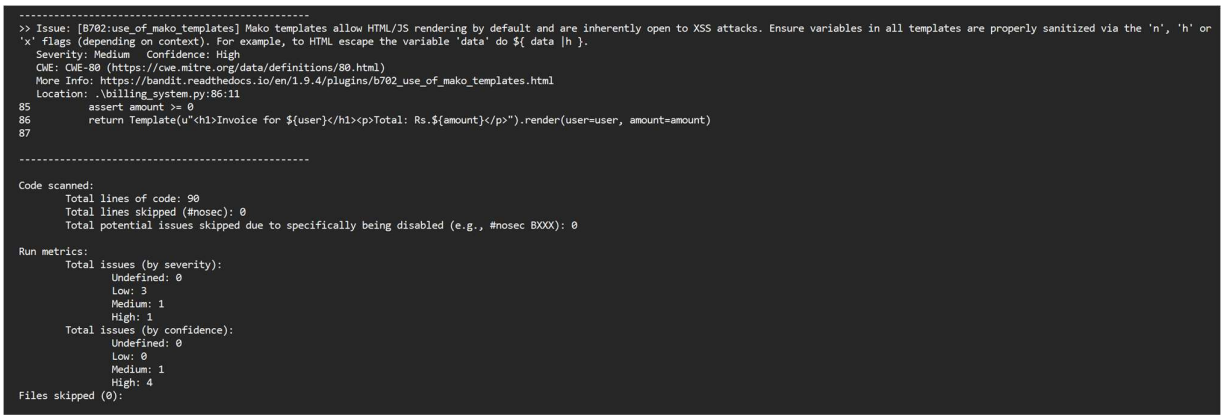
```
Run started:2026-10-30 02:46:17.560891+00:00

Test results:
>> Issue: [B413:blacklist] The pyCrypto library and its module AES are no longer actively maintained and have been deprecated. Consider using pyca/cryptography library.
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b413-import-pycrypto
   Location: .\billing_system.py:8:0
7       from mako.template import Template
8       from Crypto.Cipher import AES
9

--------------------------------------------------
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'whsec_live_51HxQ2bK'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\billing_system.py:14:17
13      MAX_ATTEMPTS = 3
14      WEBHOOK_SECRET = "whsec_live_51HxQ2bK"
15      CARD_KEY = b"billing-key-1234"

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\billing_system.py:67:29
66      def couponCode():
67          return "SAVE-" + "".join(random.choice("ABCDEFGHJKLMNPQRSTUVWXYZ23456789") for _ in range(6))
68

--------------------------------------------------
>> Issue: [B101:assert_used] Use of assert detected. The enclosed code will be removed when compiling to optimised byte code.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b101_assert_used.html
   Location: .\billing_system.py:85:4
84      def renderInvoice(user, amount):
85          assert amount >= 0
86          return Template(u"<h1>Invoice for ${user}</h1><p>Total: Rs.${amount}</p>").render(user=user, amount=amount)
```



```
--------------------------------------------------
>> Issue: [B702:use_of_mako_templates] Mako templates allow HTML/JS rendering by default and are inherently open to XSS attacks. Ensure variables in all templates are properly sanitized via the 'n', 'h' or
'x' flags (depending on context). For example, to HTML escape the variable 'data' do ${ data |h }.
   Severity: Medium   Confidence: High
   CWE: CWE-80 (https://cwe.mitre.org/data/definitions/80.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b702_use_of_mako_templates.html
   Location: .\billing_system.py:86:11
85          assert amount >= 0
86          return Template(u"<h1>Invoice for ${user}</h1><p>Total: Rs.${amount}</p>").render(user=user, amount=amount)
87

--------------------------------------------------

Code scanned:
        Total lines of code: 90
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 3
                Medium: 1
                High: 1
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 4
Files skipped (0):
```

2) pylint_report.txt

```
************* Module billing_system
billing_system.py:86:0: C0301: Line too long (111/100) (line-too-long)
billing_system.py:96:0: C0301: Line too long (126/100) (line-too-long)
billing_system.py:98:0: C0301: Line too long (103/100) (line-too-long)
billing_system.py:100:0: C0301: Line too long (108/100) (line-too-long)
billing_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
billing_system.py:1:0: C0410: Multiple imports on one line (os, sys) (multiple-imports)
billing_system.py:16:16: R1735: Consider using '{}' instead of a call to 'dict'. (use-dict-literal)
billing_system.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:21:88: R1734: Consider using [] instead of list() (use-list-literal)
billing_system.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:24:0: C0103: Function name "isTrialOver" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:27:0: C0103: Function name "applyCoupon" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:29:7: C0121: Comparison 'coupon == None' should be 'coupon is None' (singleton-comparison)
billing_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:35:0: C0103: Function name "invoiceAmount" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:40:0: C0103: Function name "nextBillingDate" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:43:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:43:0: C0103: Function name "prorateUpgrade" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:50:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:50:0: C0103: Function name "retryPayment" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:58:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:58:0: C0103: Function name "cancelRefund" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:63:8: W0134: 'return' shadowed by the 'finally' clause. (return-in-finally)
billing_system.py:63:8: W0150: return statement in finally block may swallow exception (lost-exception)
billing_system.py:63:15: E0601: Using variable 'refund' before assignment (used-before-assignment)
billing_system.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:66:0: C0103: Function name "couponCode" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:69:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:69:0: C0103: Function name "activeUsers" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:70:11: R1718: Consider using a set comprehension (consider-using-set-comprehension)
billing_system.py:70:0: C0206: Consider iterating with .items() (consider-using-dict-items)
billing_system.py:72:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:72:0: C0103: Function name "encryptCard" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:76:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:76:0: C0103: Function name "verifyWebhook" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:80:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:80:0: C0103: Function name "parseWebhook" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:81:11: I1101: Module 'lxml.etree' has no 'fromstring' member, but source is unavailable. Consider adding this module to extension-pkg-allow-list if you want to perform analysis based on run-
time introspection of living objects. (c-extension-no-member)
billing_system.py:84:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:84:0: C0103: Function name "renderInvoice" doesn't conform to snake_case naming style (invalid-name)
billing_system.py:86:20: W1406: The u prefix for strings is no longer necessary in Python >=3.0 (redundant-u-string-prefix)
billing_system.py:88:0: C0116: Missing function or method docstring (missing-function-docstring)
billing_system.py:1:0: W0611: Unused import os (unused-import)
billing_system.py:1:0: W0611: Unused import sys (unused-import)

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Your code has been rated at 4.43/10
```

3) flake8_report.txt

```
billing_system.py:1:1: F401 'os' imported but unused
billing_system.py:1:1: F401 'sys' imported but unused
billing_system.py:1:10: E401 multiple imports on one line
billing_system.py:20:1: E302 expected 2 blank lines, found 1
billing_system.py:21:80: E501 line too long (95 > 79 characters)
billing_system.py:24:1: E302 expected 2 blank lines, found 1
billing_system.py:27:1: E302 expected 2 blank lines, found 1
billing_system.py:29:15: E711 comparison to None should be 'if cond is None:'
billing_system.py:35:1: E302 expected 2 blank lines, found 1
billing_system.py:40:1: E302 expected 2 blank lines, found 1
billing_system.py:43:1: E302 expected 2 blank lines, found 1
billing_system.py:50:1: E302 expected 2 blank lines, found 1
billing_system.py:58:1: E302 expected 2 blank lines, found 1
billing_system.py:66:1: E302 expected 2 blank lines, found 1
billing_system.py:67:80: E501 line too long (97 > 79 characters)
billing_system.py:69:1: E302 expected 2 blank lines, found 1
billing_system.py:70:80: E501 line too long (80 > 79 characters)
billing_system.py:72:1: E302 expected 2 blank lines, found 1
billing_system.py:76:1: E302 expected 2 blank lines, found 1
billing_system.py:77:80: E501 line too long (82 > 79 characters)
billing_system.py:80:1: E302 expected 2 blank lines, found 1
billing_system.py:84:1: E302 expected 2 blank lines, found 1
billing_system.py:86:80: E501 line too long (111 > 79 characters)
billing_system.py:88:1: E302 expected 2 blank lines, found 1
billing_system.py:91:80: E501 line too long (94 > 79 characters)
billing_system.py:92:80: E501 line too long (95 > 79 characters)
billing_system.py:93:80: E501 line too long (97 > 79 characters)
billing_system.py:94:80: E501 line too long (94 > 79 characters)
billing_system.py:95:80: E501 line too long (80 > 79 characters)
billing_system.py:96:80: E501 line too long (126 > 79 characters)
billing_system.py:98:80: E501 line too long (103 > 79 characters)
billing_system.py:100:80: E501 line too long (108 > 79 characters)
billing_system.py:101:80: E501 line too long (80 > 79 characters)
billing_system.py:103:80: E501 line too long (89 > 79 characters)
billing_system.py:104:80: E501 line too long (82 > 79 characters)
billing_system.py:107:1: E305 expected 2 blank lines after class or function definition, found 1
```

After:

1) bandit_report.txt

```
Run started:2026-10-30 02:48:16.835174+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 143
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

2) pylint_report.txt

```
------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

3) flake8_report.txt

